

```

/*Siddiqa Bagwan B24CE1093 SYBtech2*/
#include <iostream>
using namespace std;

// Function to calculate factorial
int factorial(int n) {
    int fact = 1;
    for (int i = 1; i <= n; i++) {
        fact *= i;
    }
    return fact;
}

// Function to calculate permutation nPr
int nPr(int n, int r) {
    return factorial(n) / factorial(n - r);
}

int main() {
    int n = 12; // total participants
    int r = 5; // top 5 positions
    // Normal permutation P(12,5)
    int perm = nPr(n, r);
    // Divide by 2! because last 2 positions are identical
    int result = perm / factorial(2);
    cout << "Number of ways = " << result << endl;
    return 0;
}

```

Output

Number of ways = 47520

=== Code Execution Successful ===

```
#include <iostream>
using namespace std;

// Function to calculate factorial
long long factorial(int n) {
    long long fact = 1;
    for (int i = 1; i <= n; i++) {
        fact *= i;
    }
    return fact;
}

// Function for nCr (combinations)
long long nCr(int n, int r) {
    return factorial(n) / (factorial(r) * factorial(n - r));
}

int main() {
    int n = 20; // total students
    int r = 3; // team size
    long long result = nCr(n, r);
    cout << "Number of different teams of 3 students = " << result << endl;
    return 0;
}
```

Output

```
Number of different teams of 3 students = 1140
```

```
=== Code Execution Successful ===
```